

CSCWD: Cross-Scale Channel-wise Knowledge Distillation for Lightweight Tiny Object Detection on Edge Devices

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Abstract

Real-time tiny object detection in aerial imagery is constrained by the weak spatial evidence of very small objects and the loss of high-resolution detail in lightweight detectors. This study presents Cross-Scale Channel-wise Knowledge Distillation (CSCWD), a training-time framework that transfers high-resolution spatial representations from a YOLO11m-P2 teacher to a compact YOLO11n student without altering the student’s inference architecture. Unlike conventional same-scale feature distillation, CSCWD transfers supervision from teacher P_2 to student P_3 after feature alignment while retaining same-scale distillation at deeper pyramid levels. Under the unified seven-sequence Drone-vs-Bird validation protocol, YOLO11n-CSCWD achieves 50.17% mean average precision at an intersection-over-union threshold of 0.5 (mAP@0.5) and 59.73% recall, improving the matched CA-YOLO11n baseline by 2.92 percentage points in mAP@0.5 and 3.55 points in recall. Cross-scale alignment further increases mAP@0.5 by 2.09 points over the corresponding same-scale channel-wise distillation configuration. In zero-shot evaluation on DUT-Anti-UAV, mAP@0.5 increases from 48.29% to 50.06% without target-domain fine-tuning. This domain was included because its challenging small targets make low-latency, computationally efficient detection particularly relevant. On Raspberry Pi 5 using NCNN-FP16 at 640×640 resolution, the 2.58-million-parameter student achieves 50.32% mAP@0.5 at 82.32 ms mean wall-clock latency, or 12.15 frames per second, while retaining essentially the same runtime and memory requirements as the matched baseline. The results support cross-scale distillation for improving tiny-target detection without increasing inference-time model complexity.

Keywords: Tiny object detection, Knowledge distillation, Cross-scale feature distillation, Lightweight object detection, Edge computing, Real-time image processing

1 Introduction

Tiny object detection is difficult because targets spanning only a few pixels provide limited appearance information and weak localization evidence

[1]. In long-range visual detection of small objects, problems such as imaging distance, scale variation, motion blur, low contrast, and background clutter further weaken these cues. Long-range visual detection of unmanned aerial vehicles (UAVs) is

a representative instance of this problem because targets frequently occupy tiny image regions while practical detectors operate under latency, memory, and computational constraints on embedded hardware [2, 3]. Accordingly, this study evaluates the proposed approach on UAV imagery, where preserving fine spatial evidence under real-time edge constraints is relevant.

Existing approaches improve tiny-target representation through multi-scale processing, higher-capacity backbones, and temporal compensation. Representative Drone-vs-Bird methods use multi-scale inference, wide residual networks, or tracking-assisted recovery of missed detections [4–6]. These strategies can strengthen weak-target representation, but their additional computation and memory demand can increase inference latency on low-power platforms [7]. Lightweight detectors reduce inference complexity, yet the accompanying loss of feature resolution and representational capacity can discard shallow high-resolution cues needed for tiny-target localization. This accuracy–efficiency tension motivates training-time mechanisms that enrich a compact detector without enlarging its network.

Knowledge distillation (KD) transfers information from a high-capacity teacher to a compact student during training [8]. Feature-level KD is particularly relevant to detection because intermediate maps retain spatial structure that final predictions do not preserve. Channel-wise knowledge distillation (CWD) matches channel-wise spatial distributions, but conventional scale-matched alignment cannot directly exploit a high-resolution teacher feature level when the student has no corresponding pyramid level [9, 10]. For tiny objects, this mismatch restricts the transfer of localization-sensitive spatial detail, motivating cross-scale feature distillation.

This work presents Cross-Scale Channel-wise Knowledge Distillation (CSCWD) using a YOLO11m-P2 teacher with a high-resolution P_2 detection branch and a standard YOLO11n student [11]. CSCWD transfers supervision from teacher P_2 to student P_3 while retaining matched-scale distillation at P_4 and P_5 . Teacher and auxiliary distillation components are removed after training, leaving the student inference graph unchanged. The resulting student is evaluated

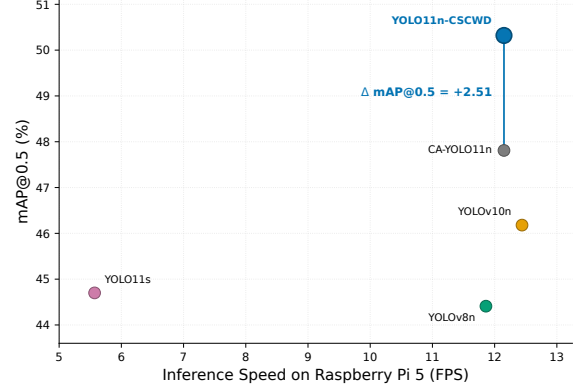


Fig. 1: Deployed accuracy and inference speed of lightweight detectors on Raspberry Pi 5 using NCNN-FP16 at 640×640 under the unified seven-sequence validation protocol

against CA-YOLO11n [12] and lightweight detectors on Raspberry Pi 5; Fig. 1 summarizes deployed accuracy and inference speed.

The main contributions are summarized as follows:

- A cross-scale CWD formulation transfers high-resolution teacher knowledge from P_2 to student P_3 while retaining deeper same-scale distillation, targeting the loss of fine spatial information in lightweight tiny-object detection.
- A deployment-preserving teacher–student formulation restricts high-resolution supervision to training, allowing the distilled model to retain the original YOLO11n inference graph and computational footprint.
- Controlled ablations, Drone-vs-Bird evaluation, zero-shot DUT-Anti-UAV testing, and physical Raspberry Pi 5 deployment quantify detection gains, zero-shot cross-dataset performance, and embedded inference efficiency [13, 14].

2 Related Work

Prior work relevant to lightweight tiny-object detection has sought to strengthen weak-target representation through multi-scale inference, increased feature-extraction capacity, and temporal compensation. On Drone-vs-Bird, Laroca et al. combine multi-scale processing with data augmentation and temporal post-processing, Wong et al. employ a WideResNet-based detector, and Sawada et al. recover missed detections

using tracking-assisted temporal information [4–6]. These approaches improve detection performance through different mechanisms but add computation or pipeline complexity, limiting their suitability when deployment complexity must remain low. Context-aware augmentation offers a training-only alternative that improves a nano-scale detector without modifying its inference graph [12]; this augmentation strategy, however, does not directly provide teacher-derived feature supervision.

Building on the original KD formulation [15], feature-level methods include intermediate representation regression, attention transfer, masked generative distillation, and focal-and-global knowledge distillation [16–19]. These methods differ in the form of supervision, but they share the goal of improving the student through training-time representation transfer without adding inference modules. CWD represents feature knowledge through channel-wise spatial probability distributions, making channel-specific spatial structure explicit during transfer [9]. Many feature-level formulations still rely on matching selected teacher and student representations at compatible or corresponding scales. ScaleKD specifically addresses scale-aware distillation for small-object detection through a scale-decoupled feature distillation module and a cross-scale assistant [10]. A narrower architectural gap arises when the teacher contains a high-resolution P_2 level while a lightweight student begins at P_3 , leaving no direct same-scale counterpart for teacher P_2 . Transferring this teacher-only high-resolution representation while preserving the student’s original inference graph motivates the proposed cross-scale P_2 -to- P_3 transfer.

3 Proposed Method

The proposed CSCWD framework transfers teacher-only high-resolution representations to a lightweight student during training while preserving the student’s original inference architecture.

3.1 Framework Overview

CSCWD uses an asymmetric teacher–student configuration to transfer high-resolution supervision that is unavailable in the student’s native feature

pyramid. During training, the frozen YOLO11m- P_2 teacher provides cross-scale P_2 -to- P_3 supervision together with matched-scale distillation at P_4 and P_5 , while the YOLO11n student remains the only network retained for deployment. Figure 2 summarizes the complete training and inference flow.

3.2 Asymmetric Teacher–Student Design

The teacher and student are deliberately asymmetric in model capacity and feature resolution. The teacher extends YOLO11m [11] with a high-resolution P_2 detection branch, producing a 160×160 feature map for a 640×640 input. The teacher is trained in two stages: VisDrone [21] adaptation followed by target-domain fine-tuning. The corresponding optimization and augmentation settings are summarized in Table 1.

The student retains the standard YOLO11n architecture without a P_2 branch and therefore begins its detection pyramid at P_3 . It is initialized from COCO-pretrained weights and adapted on VisDrone before CSCWD training. This asymmetry preserves the lightweight student graph while exposing it during training to spatial information available only at the teacher’s higher-resolution P_2 level.

3.3 Cross-Scale Feature Alignment

CSCWD transfers supervision from teacher P_2 to student P_3 , while P_4 and P_5 are distilled at matched scales. For loss computation, the student representations are aligned to the corresponding

Table 1: Teacher training settings for VisDrone adaptation and target fine-tuning.

Setting	Stage 1: VisDrone adaptation	Stage 2: Target fine-tuning
Initialization	COCO (yolo11m.pt)	Stage-1 best
Epochs / batch	30 / 4	100 / 12
Optimizer	Default	SGD
Mosaic (p /close)	1.0 / last 5	0.75 / last 10
MixUp	0.0	0.15
Scale / translate	0.5 / 0.1	0.5 / 0.0
Flip (V/H)	0.5 / 0.5	0.0 / 0.0
HSV (H/S/V)	0.015 / 0.7 / 0.4	0.0 / 0.0 / 0.4

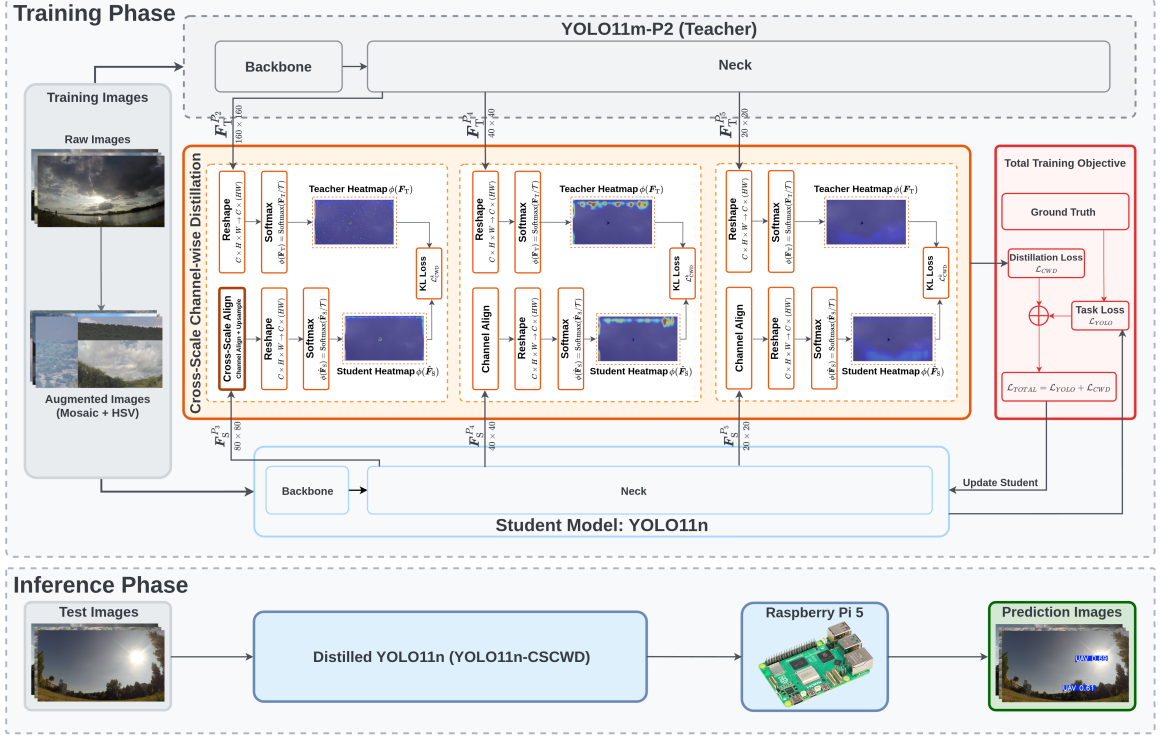


Fig. 2: Overview of CSCWD training and inference. The frozen YOLO11m-P2 teacher provides cross-scale P_2 -to- P_3 supervision and matched-scale distillation at P_4 and P_5 during training. Inference retains only the distilled YOLO11n-CSCWD model. Raspberry Pi image: CC BY 2.0 [20]

teacher feature spaces:

$$S:P_3 \rightarrow T:P_2, \quad S:P_4 \rightarrow T:P_4, \quad S:P_5 \rightarrow T:P_5 \quad (1)$$

The first pair defines the cross-scale transfer, whereas the remaining two pairs are scale matched. For loss computation, each student representation is projected to the channel dimension of its paired teacher feature. The $S:P_3$ feature is additionally upsampled to the spatial resolution of $T:P_2$; P_4 and P_5 require channel projection only.

3.4 Cross-Scale Channel-wise Knowledge Distillation

To transfer the aligned feature representations, the framework extends Channel-wise Distillation (CWD) [9] to support cross-scale matching. Unlike feature-regression approaches such as FitNets [16], CWD converts each feature channel into a spatial probability distribution and minimizes the Kullback–Leibler (KL) divergence between the teacher and student distributions.

Let T and S denote the frozen teacher and trainable student networks, respectively. For each feature pair defined by these alignment mappings $P_3 \rightarrow P_2$, $P_4 \rightarrow P_4$, and $P_5 \rightarrow P_5$, the intermediate feature representations are denoted as:

$$\mathbf{F}_T^k \in \mathbb{R}^{N \times C_T^k \times H_T^k \times W_T^k} \quad (2)$$

and

$$\mathbf{F}_S^k \in \mathbb{R}^{N \times C_S^k \times H_S^k \times W_S^k}, \quad (3)$$

where N represents the batch size, and C , H , and W denote the channel, height, and width dimensions of the feature tensors. For the cross-scale pair ($P_3 \rightarrow P_2$), $\mathbf{F}_T^k$ specifically represents the teacher’s high-resolution P_2 feature.

Channel alignment and spatial matching

Since teacher and student features have different channel dimensions, the student representation

is first projected into the teacher feature space using a learnable 1×1 convolution. The channel transformation function is defined as:

$$g_{\text{map}}^k : \mathbb{R}^{N \times C_S^k \times H_S^k \times W_S^k} \longrightarrow \mathbb{R}^{N \times C_T^k \times H_S^k \times W_S^k}. \quad (4)$$

The channel-aligned student feature is obtained by:

$$\tilde{\mathbf{F}}_S^k = g_{\text{map}}^k(\mathbf{F}_S^k). \quad (5)$$

If the spatial resolution does not match the corresponding teacher feature, bilinear interpolation is applied:

$$\text{Upsample} : \mathbb{R}^{N \times C_T^k \times H_S^k \times W_S^k} \longrightarrow \mathbb{R}^{N \times C_T^k \times H_T^k \times W_T^k}. \quad (6)$$

Therefore, the final aligned student feature is obtained as:

$$\hat{\mathbf{F}}_S^k = \begin{cases} \text{Upsample}(\tilde{\mathbf{F}}_S^k; H_T^k, W_T^k), & \text{if } (H_S^k, W_S^k) \neq (H_T^k, W_T^k), \\ \tilde{\mathbf{F}}_S^k, & \text{otherwise.} \end{cases} \quad (7)$$

After alignment, teacher and student features share identical dimensions and can be compared through the distillation objective.

Channel-wise probability distributions

Following the CWD formulation, each feature channel is normalized independently over spatial positions using a temperature-scaled softmax. Let $x_{c,i}^k$ denote the activation value of channel c at spatial position i . The spatial probability distribution is defined as:

$$\phi(x_{c,i}^k) = \frac{\exp(x_{c,i}^k/\mathcal{T})}{\sum_{j=1}^{H^k W^k} \exp(x_{c,j}^k/\mathcal{T})}, \quad (8)$$

where $\mathcal{T}$ denotes the temperature parameter. This normalization converts each channel activation map into a spatial probability distribution, allowing the student to learn the teacher's spatial attention patterns.

Channel-wise distillation loss

The distillation loss is calculated by minimizing the KL divergence between the teacher and aligned student channel distributions:

$$\mathcal{L}_{\text{CWD}}^k = \frac{\mathcal{T}^2}{C^k} \sum_{c=1}^{C^k} \sum_{i=1}^{H^k W^k} \phi(\mathbf{F}_{T,c,i}^k) \log \left[\frac{\phi(\mathbf{F}_{T,c,i}^k)}{\phi(\hat{\mathbf{F}}_{S,c,i}^k)} \right] \quad (9)$$

where C^k represents the number of channels at the corresponding aligned feature level. The complete distillation objective is obtained by aggregating all feature mappings:

$$\mathcal{L}_{\text{CWD}} = \alpha_{\text{CWD}} \sum_{k \in \{P_3, P_4, P_5\}} \mathcal{L}_{\text{CWD}}^k, \quad (10)$$

where α_{CWD} controls the contribution of the distillation term.

Total training objective

The final optimization objective combines the original detection loss and the proposed distillation loss:

$$\mathcal{L}_{\text{TOTAL}} = \mathcal{L}_{\text{YOLO}} + \mathcal{L}_{\text{CWD}}. \quad (11)$$

All teacher, projection, and spatial-alignment components are discarded after training. Deployment therefore retains only the distilled YOLO11n student (YOLO11n-CSCWD), preserving the original inference graph and computational footprint.

4 Experiments and Results

Experiments evaluate CSCWD under a unified Drone-vs-Bird protocol, zero-shot cross-dataset testing, controlled ablations, and physical deployment on Raspberry Pi 5.

4.1 Experimental Setup and Evaluation Protocol

Datasets and evaluation. Drone-vs-Bird [2, 13] is the primary benchmark. Frames are sampled at 2 FPS, and the seven-sequence validation protocol of Laroca et al. [4] contains 12,381 labeled frames. The seven sequences form the validation split; the reported metrics are computed separately for each sequence and averaged with equal weight, while the remaining 70 of 77 sequences are

used for target-domain training. Zero-shot evaluation uses the 2,200-image DUT-Anti-UAV test split [14], without target-domain fine-tuning.

Training configuration. CA-YOLO11n denotes the matched baseline trained with the context-aware augmentation strategy of Zamani et al. [12]. Before distillation, the student is initialized from COCO-pretrained YOLO11n weights and adapted to VisDrone [21] for 50 epochs. This aerial-domain adaptation uses Mosaic with probability 1.0, vertical and horizontal flips with probability 0.5, and HSV color augmentation. CSCWD training then proceeds for 100 epochs on a single NVIDIA RTX 5070 Ti GPU with 16 GB VRAM using SGD (batch size 32, momentum 0.937, weight decay 5×10^{-4}), unless a specific ablation variant is stated otherwise. The final distillation configuration uses $\mathcal{T} = 4$ and $\alpha_{\text{CWD}} = 150$.

Metrics and deployment protocol. Detection performance is reported using Precision, Recall, mAP@0.5, and mAP@0.5:0.95. Desktop PyTorch and deployed NCNN-FP16 accuracies are reported separately. For intermediate-feature analysis, a normalized Target Concentration Ratio (nTCR) is defined as a diagnostic of spatial selectivity. Here, $z_c(u)$ denotes the activation of channel c at spatial position u , C is the channel count, H and W are feature-map dimensions, v spans all spatial positions, Ω_{GT} denotes the ground-truth region, and $q(u)$ is the channel-averaged spatial probability.

$$\text{nTCR} = \frac{\sum_{u \in \Omega_{\text{GT}}} q(u)}{|\Omega_{\text{GT}}|/(HW)}, \quad q(u) = \frac{1}{C} \sum_{c=1}^C \frac{\exp(z_c(u)/\mathcal{T})}{\sum_v \exp(z_c(v)/\mathcal{T})} \quad (12)$$

Values above 1 indicate target-region enrichment relative to a spatially uniform allocation. The matched comparison uses CA-YOLO11n and YOLO11n-CSCWD at P_3 over the 11,128 validation frames containing at least one ground-truth target; teacher P_2 is descriptive only because of its different spatial resolution.

Raspberry Pi 5 measurements use the 8 GB model, a four-core 2.4 GHz ARM CPU, Debian 13, NCNN-FP16, batch size 1, and the `performance` governor. CA-YOLO11n and YOLO11n-CSCWD use a 10-image warm-up and three full validation passes at 640 and 960; 1280 is accuracy-only. Figure 1 reference-detector FPS values use one full 640 pass on the same hardware and backend.

Mean wall-clock latency, 95th-percentile (P95) latency, FPS, and peak resident set size (RSS) are reported.

4.2 Benchmark Comparison on Drone-vs-Bird

Controlled comparisons examine whether the CSCWD gain is retained against representative lightweight YOLO detectors under the same Drone-vs-Bird training and evaluation setting. YOLO11n-CSCWD is compared with YOLOv8n, YOLOv10n, YOLO11s, the matched CA-YOLO11n baseline [12], and the YOLO11m-P2 teacher. The controlled detectors use family-specific VisDrone-adapted initialization and otherwise follow the same Drone-vs-Bird training configuration. Table 2 summarizes these controlled results together with contextual values reported in the literature.

Under the unified seven-sequence validation protocol, YOLO11n-CSCWD improves CA-YOLO11n by 2.92 percentage points in mAP@0.5 and achieves 4.00 points higher mAP@0.5 than the larger YOLO11s model, while retaining the 2.58 M-parameter, 6.4-GFLOP YOLO11n inference graph. YOLO11m-P2 reaches 56.75% mAP@0.5 but requires 20.82 M parameters and 134.5 GFLOPs; this higher-capacity teacher is used only during training. Literature-reported values are treated as contextual references rather than direct rankings because their data splits and inference procedures differ.

4.3 Sequence-Level and Cross-Dataset Analysis

Sequence-level results show that YOLO11n-CSCWD outperforms CA-YOLO11n on six of the seven sequences, with the largest gains on Swarm (+10.34 points) and Hillside (+9.23 points). GoPro 002 is the only sequence showing a decrease, from 28.86% to 23.48% mAP@0.5, indicating scenario-dependent behavior (Table 3).

Target scale alone does not explain sequence difficulty. Inspire contains comparatively larger resized targets but remains difficult for the controlled detectors, whereas GoPro 002 combines extremely small targets with the clearest CSCWD failure case. The reduced qualitative comparison

Table 2: Comparison on Drone-vs-Bird. The first three rows report literature values obtained under different data splits or inference procedures; the remaining models are evaluated under the unified seven-sequence validation protocol.

Method	Architecture	Split type	Params (M)	GFLOPs	mAP @0.5	mAP @0.5:0.95
Wong et al. [5]	WideResNet-based	Frame-based	100.70	260.0	75.70 ^b	–
Sawada et al. [6]	Faster R-CNN + RFLA	Video-based	23.50	130.0	64.18 ^a	–
Laroca et al. [4]	YOLO11m (whole image)	Video-based	20.10	68.0	55.23 ^c	–
YOLOv8n [22]	YOLOv8n	Video-based	3.20	8.7	45.74	15.90
YOLOv10n [23]	YOLOv10n	Video-based	2.30	6.7	44.49	15.42
YOLO11s [11]	YOLO11s	Video-based	9.40	21.6	46.17	16.37
CA-YOLO11n [12]	YOLO11n	Video-based	2.58	6.4	47.25	16.92
YOLO11n-CSCWD	YOLO11n	Video-based	2.58	6.4	50.17	17.30
YOLO11m-P2	YOLO11m + P2 branch	Video-based	20.82	134.5	56.75	21.26

^aDifferent seven-video split in the original study. ^bIntra-video frame-based random split. ^cWhole-image-only result; the complete multi-scale and temporal pipeline reports 73.90%.

in Fig. 3 therefore illustrates successful detections under two challenging visual conditions.

Zero-shot evaluation on DUT-Anti-UAV is reported in Table 4. The Drone-vs-Bird-trained checkpoints are applied directly to the DUT test split without target-domain fine-tuning. YOLO11n-CSCWD improves CA-YOLO11n by 3.98 points in precision, 1.61 points in recall, and 1.77 points in mAP@0.5, whereas mAP@0.5:0.95 remains essentially unchanged (24.00% versus 24.04%).

4.4 Ablation Analysis

Two controlled ablations isolate the effects of the feature-distillation objective and teacher–student feature pairing. The first compares alternative

feature-transfer objectives, whereas the second separates teacher design from cross-scale alignment. Both ablations retain the same student initialization and data pipeline unless explicitly stated otherwise.

4.4.1 Distillation Objective

CWD [9] is compared with Masked Generative Distillation (MGD) [18] and a hybrid objective combining $\mathcal{L}_{\text{YOLO}}$, $\mathcal{L}_{\text{CWD}}$, and $\mathcal{L}_{\text{MGD}}$.

Among the evaluated objectives, MGD achieves the highest mAP@0.5 (50.53%) and mAP@0.5:0.95 (18.00%), whereas YOLO11n-CSCWD provides the highest Recall (59.73%) with 50.17% mAP@0.5 (Table 5). The 0.36-point mAP@0.5 difference does not support a claim of universal CWD superiority; the principal CSCWD contribution is instead the cross-scale feature pairing examined next.

Table 3: Sequence-level mAP@0.5 under the unified seven-sequence validation protocol.

Sequence	CA-YOLO11n	YOLO11n-CSCWD	YOLO11m-P2
Mavic	91.28	91.41	93.07
Inspire	31.86	35.99	36.56
Disco	79.33	79.48	84.38
GOPR5843	26.80	28.63	37.73
Swarm	48.98	59.32	56.65
Hillside	23.61	32.84	54.96
GoPro 002	28.86	23.48	33.87

Table 4: Zero-shot DUT-Anti-UAV evaluation without target-domain fine-tuning.

Model	Precision	Recall	mAP @0.5	mAP @0.5:0.95
CA-YOLO11n	70.61	46.59	48.29	24.04
YOLO11n-CSCWD	74.59	48.20	50.06	24.00
YOLO11m-P2	83.33	56.34	62.57	30.43

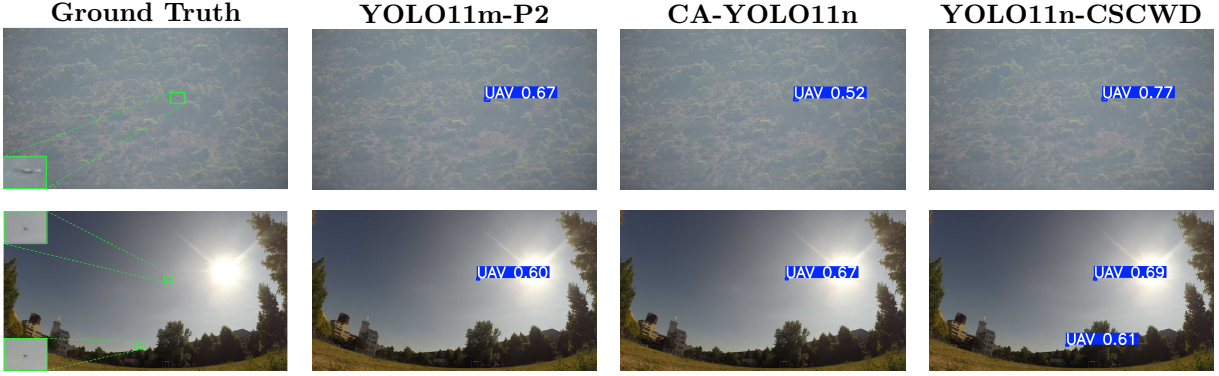


Fig. 3: Qualitative Drone-vs-Bird examples comparing ground truth, YOLO11m-P2, CA-YOLO11n, and YOLO11n-CSCWD. The selected cases illustrate successful tiny-target detection under challenging visual conditions

Table 5: Ablation of feature-distillation objectives under the unified seven-sequence validation protocol.

Model	Precision	Recall	mAP @0.5	mAP @0.5:0.95
CA-YOLO11n [12]	61.72	56.18	47.25	16.92
YOLO11n-MGD [18]	63.96	58.71	50.53	18.00
YOLO11n-Hybrid	63.68	57.16	49.70	17.36
YOLO11n-CSCWD	63.42	59.73	50.17	17.30

4.4.2 Teacher Design and Cross-Scale Alignment

The second ablation distinguishes the contribution of the P_2 -enhanced teacher from that of cross-scale teacher-student pairing.

Adding P_2 increases teacher mAP@0.5 from 52.84% to 56.75%, whereas CWD using the teacher without P_2 yields only 44.08% for the student. With the P_2 -enhanced teacher, same-scale CWD reaches 48.08%, while YOLO11n-CSCWD reaches 50.17% and increases Recall from 55.36% to 59.73%, despite a modest reduction in Precision (Table 6). The 2.09-point mAP@0.5 gain over same-scale CWD indicates that the student improvement is not explained by teacher capacity alone and also depends on the proposed cross-scale pairing.

4.5 Embedded Deployment on Raspberry Pi 5

Raspberry Pi 5 results in Table 7 report NCNN-FP16 accuracy and application-level runtime at

640 and 960, while 1280 remains an accuracy-only setting.

At 640×640 , YOLO11n-CSCWD raises deployed mAP@0.5 from 47.81% to 50.32% while both models remain at approximately 82.3 ms mean wall-clock latency and 12.15 FPS. At 960×960 , the mAP@0.5 advantage widens to 4.86 points (52.85% versus 47.99%) while latency and throughput again remain essentially unchanged. Peak RSS is also comparable at both resolutions, consistent with the unchanged student inference graph.

Accuracy is non-monotonic with input resolution. YOLO11n-CSCWD increases from 50.32% mAP@0.5 at 640 to 52.85% at 960, but decreases to 48.53% at 1280; CA-YOLO11n shows a stronger decline. Desktop PyTorch evaluation shows the same qualitative trend at 1280 (48.17% for YOLO11n-CSCWD and 39.38% for CA-YOLO11n), making a purely NCNN-export-specific explanation unlikely. Across the measured 640 and 960 operating points, YOLO11n-CSCWD therefore improves deployed detection accuracy while retaining essentially the same latency, throughput, and memory footprint as CA-YOLO11n.

5 Discussion

The results indicate that CSCWD benefits primarily from cross-scale access to the teacher’s high-resolution representation rather than from teacher capacity or detector scaling alone.

Table 6: Effect of teacher design and feature pairing under the unified seven-sequence validation protocol.

Configuration	Precision	Recall	mAP@0.5	mAP@0.5:0.95
YOLO11m	67.66	60.52	52.84	19.17
YOLO11m-P2	69.43	62.73	56.75	21.26
YOLO11n-CWD (YOLO11m teacher)	56.21	52.45	44.08	15.58
YOLO11n-CWD (YOLO11m-P2, same-scale)	64.84	55.36	48.08	16.64
YOLO11n-CSCWD	63.42	59.73	50.17	17.30

Table 7: Raspberry Pi 5 deployment results for NCNN-FP16 artifacts. Accuracy follows the unified seven-sequence validation protocol. Runtime statistics at 640 and 960 are averaged over three complete validation runs; 1280 reports accuracy only.

Model	Input size	mAP @0.5	mAP @0.5:0.95	Mean wall latency (ms)	P95 latency (ms)	FPS	Peak RSS (MB)
CA-YOLO11n	640	47.81	16.92	82.29	109.48	12.15	561.23
	960	47.99	17.54	183.64	213.21	5.45	628.02
	1280	39.17	13.61	—	—	—	—
YOLO11n-CSCWD	640	50.32	17.36	82.32	109.97	12.15	559.81
	960	52.85	19.38	183.44	212.05	5.45	624.31
	1280	48.53	17.76	—	—	—	—

5.1 Cross-Scale Transfer and Feature Selectivity

The ablation evidence shows that a stronger teacher is not sufficient for effective student transfer. Although the P_2 branch improves the teacher, same-scale CWD with YOLO11m-P2 reaches 48.08% mAP@0.5 for the student, whereas the proposed teacher- P_2 to student- P_3 pairing reaches 50.17%. The resulting 2.09-point gain, together with the Recall increase from 55.36% to 59.73%, supports the importance of the spatial source and pairing of the transferred representation. Model scaling alone does not explain this behavior: the larger YOLO11s reference remains 4.00 points below YOLO11n-CSCWD in mAP@0.5. The loss ablation provides an important qualification. MGD gives slightly higher mAP@0.5 and mAP@0.5:0.95 than CSCWD (50.53% versus 50.17% and 18.00% versus 17.30%), while YOLO11n-CSCWD provides the highest Recall. The contribution is therefore attributed to cross-scale feature pairing rather than to universal superiority of CWD. This distinction is particularly relevant to tiny-object detection, where the location of high-resolution supervision can matter as much as the choice of transfer loss.

The nTCR analysis provides complementary evidence at the feature level. Across the 11,128 paired frames, YOLO11n-CSCWD has higher

nTCR than CA-YOLO11n on 62.7% of frames; a paired Wilcoxon signed-rank test gives $p < 0.001$. The effect is modest and sequence dependent, so nTCR is interpreted only as a diagnostic of intermediate spatial selectivity. This distinction is important because GoPro 002 shows higher nTCR for YOLO11n-CSCWD despite lower sequence-level mAP@0.5. Stronger target-region concentration is therefore not sufficient for improved final detection, where localization quality, feature resolution, detection-head behavior, and scene ambiguity also contribute. Figure 4 illustrates the same pattern qualitatively: YOLO11n-CSCWD produces a more target-focused P_3 response than CA-YOLO11n in the selected Hillside example, while the teacher P_2 response is shown only for higher-resolution comparison.

5.2 Resolution and Edge-Deployment Trade-offs

The Raspberry Pi 5 results show that input resolution acts as an empirical deployment operating point rather than a monotonic accuracy control. For YOLO11n-CSCWD, 640×640 provides the higher-throughput setting (50.32% mAP@0.5 at 12.15 FPS), whereas 960×960 provides the highest measured deployed accuracy (52.85% at 5.45 FPS). Accuracy then decreases to 48.53% at 1280×1280 . CA-YOLO11n exhibits the same

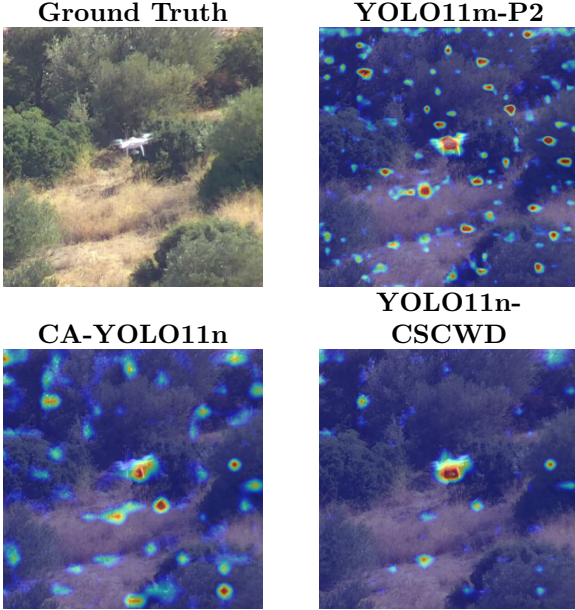


Fig. 4: Target-centered feature responses on a representative Hillside frame. YOLO11n-CSCWD shows a more concentrated P_3 response than CA-YOLO11n; YOLO11m-P2 provides a higher-resolution comparison

qualitative high-resolution degradation, and desktop PyTorch reproduces the trend, making a purely NCNN-specific export artifact unlikely to be the sole cause. Because both detectors were trained at 640, train-test scale mismatch may contribute, although the experiment does not isolate this mechanism. At a fixed resolution, the two lightweight models retain essentially identical latency, throughput, and memory usage, consistent with the unchanged YOLO11n inference graph after distillation. The resulting trade-off favors 640 when throughput is prioritized and 960 when the highest measured accuracy is more important.

The sustained-load comparison provides a secondary hardware perspective. YOLO11n-CSCWD sustains 11.85 FPS with a final-120-s mean CPU temperature of 72.49°C, whereas YOLO11m-P2 reaches 0.94 FPS and 78.13°C; no throttling is observed in either run (Fig. 5). This difference reflects deployed model scale and computational demand rather than a direct thermal effect of the distillation loss. Because one controlled 10-min run is used per model, the result is descriptive and

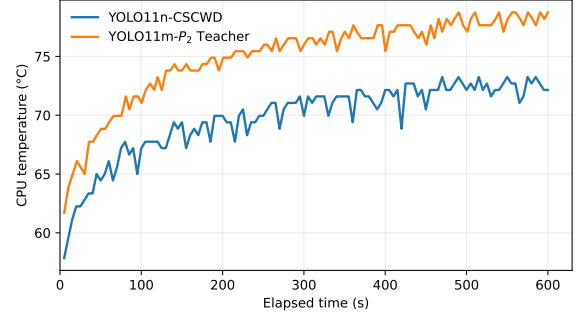


Fig. 5: Raspberry Pi 5 CPU temperature during 10 min of sustained NCNN-FP16 inference at 640×640. YOLO11n-CSCWD remains cooler than YOLO11m-P2, with no observed throttling

should not be interpreted as a statistical estimate of long-term thermal reliability.

5.3 Limitations and Future Directions

Zero-shot DUT-Anti-UAV evaluation retains an mAP@0.5 advantage for YOLO11n-CSCWD, while mAP@0.5:0.95 remains essentially unchanged. The benefit under domain shift is therefore stronger for detection confidence and sensitivity than for stricter localization quality, while a substantial gap to the higher-capacity teacher remains. This result provides cross-dataset evidence for the proposed approach but does not establish broad generalization across unseen tiny-object settings. Evaluation remains concentrated on aerial imagery, with UAVs serving as representative tiny targets. Resolution sensitivity is measured with models trained at 640 rather than with resolution-aware retraining. nTCR is correlational and does not establish a causal link between feature concentration and detection accuracy, while the thermal comparison is based on one sustained-load run per model. Future work should examine resolution-aware training, broader tiny-object benchmarks, hardware-aware compression, and lightweight temporal modeling for transient misses.

6 Conclusion

This work introduced Cross-Scale Channel-wise Knowledge Distillation (CSCWD) for lightweight tiny-object detection under resource-constrained

deployment. CSCWD transfers the teacher’s high-resolution P_2 representation to student P_3 during training while preserving the deployed YOLO11n inference graph. Using aerial UAV imagery as a representative tiny-object setting, YOLO11n-CSCWD achieves 50.17% mAP@0.5 under the unified seven-sequence Drone-vs-Bird validation protocol, improving CA-YOLO11n by 2.92 percentage points. Cross-scale P_2 -to- P_3 alignment further improves mAP@0.5 by 2.09 points over same-scale CWD, supporting the importance of teacher–student feature pairing beyond teacher capacity alone. Zero-shot DUT-Anti-UAV evaluation adds 1.77 mAP@0.5 points without target-domain fine-tuning. On Raspberry Pi 5, the NCNN-FP16 artifact achieves 50.32% mAP@0.5 at 82.3 ms mean wall-clock latency and 12.15 FPS, with runtime essentially unchanged relative to CA-YOLO11n. This separation directly preserves the intended accuracy–efficiency trade-off on embedded edge hardware. These findings support cross-scale distillation as a practical training-time strategy for improving compact tiny-object detectors without carrying teacher-side complexity into deployment.

Statements and Declarations

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Data availability. The Drone-vs-Bird dataset is available from the official challenge organizers upon request and subject to its access conditions. DUT-Anti-UAV and VisDrone are publicly available from their official repositories. Derived evaluation outputs and deployment benchmark logs are available from the corresponding author upon reasonable request, subject to the original dataset licensing conditions.

Code availability. The source code and configuration files are not publicly available at the time of this preprint release.

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